CSA4001-MANAGEMENT INFORMATION SYSTEM FOR DATA OPTIMIZATION

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10) A college has more than thousand security persons, who are instructed to give duties at different places within the campus. Additionally, they also maintain a routine, which contains all information, such as Date, Duty Start Time, Duty End Time, and Place. Most importantly, all the places are covered by at least one security person. If a security person takes leave, manual entry is done against that person. Finally, at the end of a month, the security persons get paid for their duties, while considering the number of leaves as well. You can see that the manual calculation/operation is a heavy task for the security manager. Therefore, the objective is to build an Online security management system using class diagram through which entire security system within the campus can be controlled in an efficient manner

AIM

The aim is to design a Class Diagram for an Online Security Management System to automate and streamline the management of security personnel, including their duties, leaves, and payment calculations.

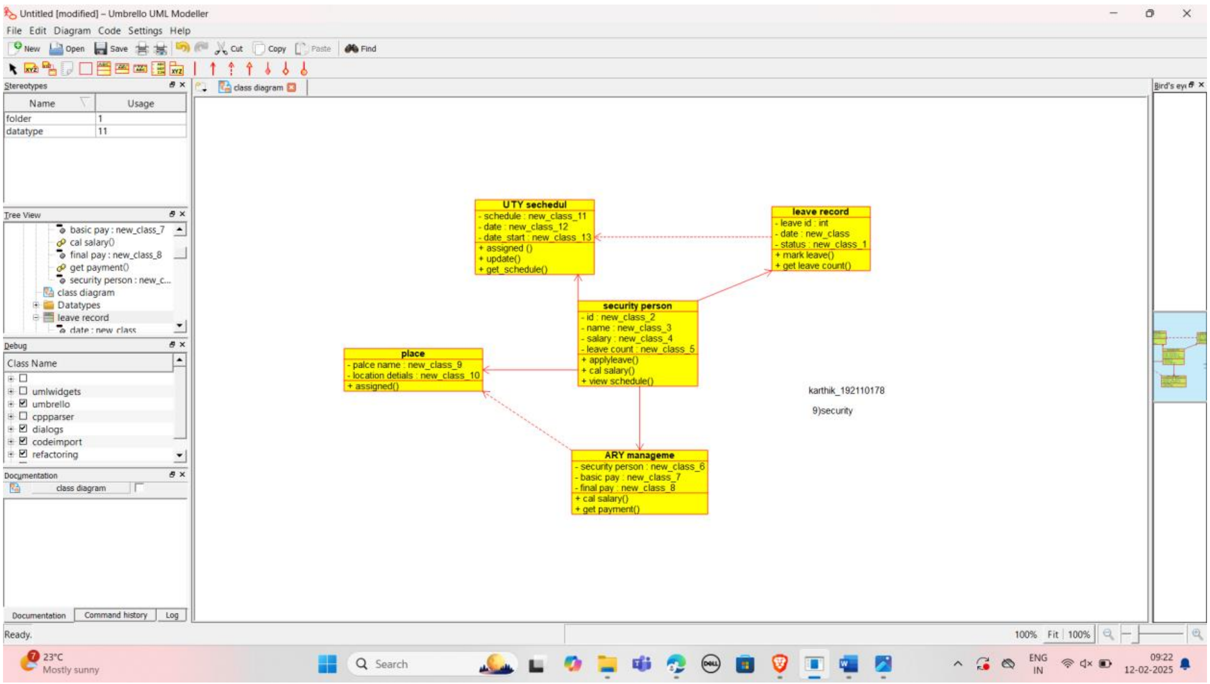
PROCEDURE:

1. Identify the main classes: SecurityPerson, DutySchedule, Leave, Payment, SecurityManager, and Place.
2. Define attributes for each class, such as personal details for SecurityPerson, duty details for DutySchedule, leave dates for Leave, and payment calculations for Payment.

3. Establish methods for each class, such as adding a duty, calculating payment, applying leave, and generating reports.
4. Define relationships between classes, such as SecurityPerson having a one-to-many relationship with DutySchedule and Leave, and SecurityManager managing all duties and payment calculations.
5. Implement inheritance where applicable, such as SecurityManager inheriting from a Person class for general attributes like name and ID.
6. Define associations, like DutySchedule linked to Place (where the security person is assigned) and SecurityPerson assigned to specific duties and leaves.
7. Finalize the diagram, ensuring all necessary functionalities like duty assignment, leave management, and payment calculation are represented.

OUTPUT:

USE CASE DIAGRAM:



RESULT:

The Class Diagram will represent the relationships between key entities (e.g., SecurityPerson, DutySchedule, Leave, Payment, etc.), providing an efficient structure for managing security duties, leave tracking, and payment processing on campus.